

# TYPES 2N3702, 2N3703, A8T3702, A8T3703

## P-N-P SILICON TRANSISTORS

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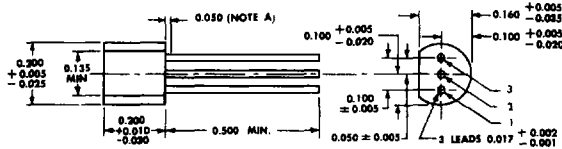
### SILECT† TRANSISTORS‡

- For Medium-Power Amplifiers, Class B Audio Outputs, Hi-Fi Drivers
- Also Available in Pin-Circle Versions . . . 2N5447, 2N5448
- For Complementary Use with 2N3704 thru 2N3706 or A8T3704 thru A8T3706

#### mechanical data

These transistors are encapsulated in a plastic compound specifically designed for this purpose, using a highly mechanized process developed by Texas Instruments. The case will withstand soldering temperatures without deformation. These devices exhibit stable characteristics under high-humidity conditions and are capable of meeting MIL-STD-202C, Method 106B. The transistors are insensitive to light.

\*ALL JEDEC TO-92 DIMENSIONS AND NOTES ARE APPLICABLE



NOTES: A. Lead diameter is not controlled in this area.  
B. All dimensions are in inches.

| DEVICE           | LEADS   |           |           |
|------------------|---------|-----------|-----------|
|                  | 1       | 2         | 3         |
| 2N3702, 2N3703   | Emitter | Collector | Base      |
| A8T3702, A8T3703 | Emitter | Base      | Collector |

2N3702  
2N3703

A8T3702  
A8T3703



ECB



EBC

#### absolute maximum ratings at 25°C free-air temperature (unless otherwise noted)

|                                                                                    | 2N3702<br>A8T3702                                                                                                                                                            | 2N3703<br>A8T3703 |
|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| Collector-Base Voltage                                                             | -40 V*                                                                                                                                                                       | -50 V*            |
| Collector-Emitter Voltage (See Note 1)                                             | -25 V*                                                                                                                                                                       | -30 V*            |
| Emitter-Base Voltage                                                               | -5 V*                                                                                                                                                                        | -5 V*             |
| Continuous Collector Current                                                       | -200 mA*                                                                                                                                                                     |                   |
| Continuous Device Dissipation at (or below) 25°C Free-Air Temperature (See Note 2) | <div style="display: flex; align-items: center;"> <div style="text-align: center;"> <div>625 mW§</div> <div>360 mW*</div> </div> <div style="margin: 0 10px;">→</div> </div> |                   |
| Continuous Device Dissipation at (or below) 25°C Lead Temperature (See Note 3)     | <div style="display: flex; align-items: center;"> <div style="text-align: center;"> <div>1.25 W§</div> <div>500 mW*</div> </div> <div style="margin: 0 10px;">→</div> </div> |                   |
| Storage Temperature Range                                                          | -65°C to 150°C*                                                                                                                                                              |                   |
| Lead Temperature 1/16 Inch from Case for 10 Seconds                                | -260°C*                                                                                                                                                                      |                   |

NOTES: 1. These values apply when the base-emitter diode is open-circuited.

2. Derate the 625-mW rating linearly to 150°C free-air temperature at the rate of 5 mW/°C. Derate the 360-mW (JEDEC registered) rating linearly to 150°C free-air temperature at the rate of 2.88 mW/°C.

3. Derate the 1.25-W rating linearly to 150°C lead temperature at the rate of 10 mW/°C. Derate the 500-mW (JEDEC registered) rating linearly to 150°C lead temperature at the rate of 4 mW/°C. Lead temperature is measured on the collector lead 1/16 inch from the case.

\*The asterisk identifies JEDEC registered data for the 2N3702 and 2N3703 only. This data sheet contains all applicable registered data in effect at the time of publication.

†Trademark of Texas Instruments

‡U.S. Patent No. 3,439,238

§Texas Instruments guarantees these values in addition to the JEDEC registered values which are also shown.

USES CHIP P20

# TYPES 2N3702, 2N3703, A8T3702, A8T3703

## P-N-P SILICON TRANSISTORS

\*electrical characteristics at 25°C free-air temperature

| PARAMETER                                             | TEST CONDITIONS                                                   | 2N3702<br>A8T3702 |     | 2N3703<br>A8T3703 |     | UNIT |
|-------------------------------------------------------|-------------------------------------------------------------------|-------------------|-----|-------------------|-----|------|
|                                                       |                                                                   | MIN               | MAX | MIN               | MAX |      |
| $V_{(BR)CBO}$ Collector-Base Breakdown Voltage        | $I_C = -100 \mu A, I_E = 0$                                       | -40               |     | -50               |     | V    |
| $V_{(BR)CEO}$ Collector-Emitter Breakdown Voltage     | $I_C = -10 \text{ mA}, I_B = 0, \text{ See Note 4}$               | -25               |     | -30               |     | V    |
| $V_{(BR)EBO}$ Emitter-Base Breakdown Voltage          | $I_E = -100 \mu A, I_C = 0$                                       | -5                |     | -5                |     | V    |
| $I_{CBO}$ Collector Cutoff Current                    | $V_{CB} = -20 \text{ V}, I_E = 0$                                 | -100              |     | -100              |     | nA   |
| $I_{EBO}$ Emitter Cutoff Current                      | $V_{EB} = -3 \text{ V}, I_C = 0$                                  | -100              |     | -100              |     | nA   |
| $h_{FE}$ Static Forward Current Transfer Ratio        | $V_{CE} = -5 \text{ V}, I_C = -50 \text{ mA}, \text{ See Note 4}$ | 60                | 300 | 30                | 150 |      |
| $V_{BE}$ Base-Emitter Voltage                         | $V_{CE} = -5 \text{ V}, I_C = -50 \text{ mA}, \text{ See Note 4}$ | -0.6              | -1  | -0.6              | -1  | V    |
| $V_{CE(sat)}$ Collector-Emitter Saturation Voltage    | $I_B = -5 \text{ mA}, I_C = -50 \text{ mA}, \text{ See Note 4}$   | -0.25             |     | -0.25             |     | V    |
| $f_T$ Transition Frequency                            | $V_{CE} = -5 \text{ V}, I_C = -50 \text{ mA}, \text{ See Note 5}$ | 100               |     | 100               |     | MHz  |
| $C_{obo}$ Common-Base Open-Circuit Output Capacitance | $V_{CB} = -10 \text{ V}, I_E = 0, f = 1 \text{ MHz}$              | 12                |     | 12                |     | pF   |

NOTES: 4. These parameters must be measured using pulse techniques.  $t_w = 300 \mu s$ , duty cycle  $\leq 2\%$ .

5. To obtain  $f_T$ , the  $|h_{fe}|$  response with frequency is extrapolated at the rate of -6 dB per octave from  $f = 20 \text{ MHz}$  to the frequency at which  $|h_{fe}| = 1$ .

\*The asterisk identifies JEDEC registered data for the 2N3702 and 2N3703 only.

### THERMAL INFORMATION

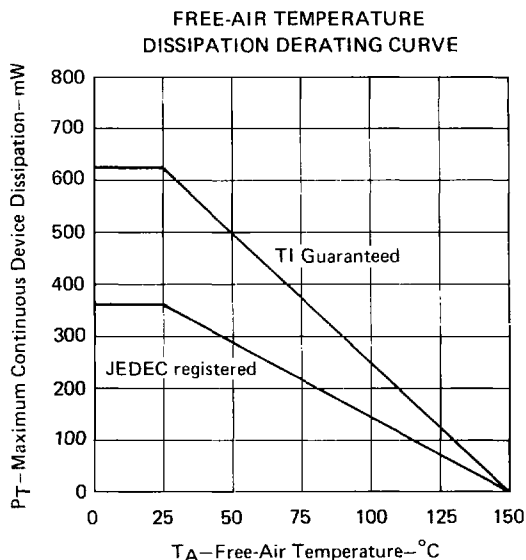


FIGURE 1

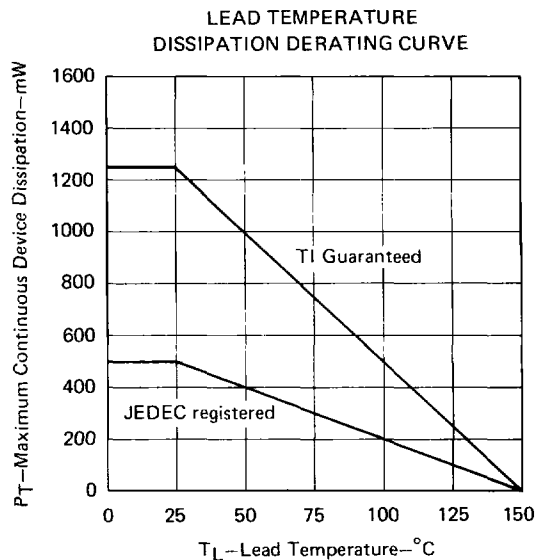


FIGURE 2